

$$C_{\text{dw}\phi_1}^{i1-i2-} \rightarrow \frac{1}{16\pi^2}$$

$$\begin{aligned} & \left(-\frac{1}{72} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_1, m_d^{i2}, m_q^{i1}] + \frac{1}{144} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_1, m_d^{i2}, m_q^{i1}] + \right. \\ & \frac{1}{72} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_1, m_d^{i2}, m_q^{i1}] - \frac{1}{144} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_1, m_q^{i1}, m_d^{i2}] - \\ & \frac{1}{24} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_1, m_q^{i1}, \tilde{\mu}] + \frac{1}{24} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_1, m_q^{i1}, \tilde{\mu}] + \\ & \frac{1}{24} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{i2}] + \frac{1}{48} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] + \\ & \frac{1}{8} m_2 \tilde{\mu} g_2^3 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_2, m_q^{i1}, \tilde{\mu}] + \frac{3}{8} m_2 \tilde{\mu} g_2^3 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_2, m_q^{i1}, \tilde{\mu}] + \\ & \frac{3}{16} m_2 \tilde{\mu} g_2^3 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] + \frac{1}{3} g_2 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_3, m_d^{i2}, m_q^{i1}] - \\ & \frac{1}{6} g_2 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_3, m_d^{i2}, m_q^{i1}] - \frac{1}{3} g_2 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_3, m_d^{i2}, m_q^{i1}] + \\ & \frac{1}{6} g_2 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_3, m_q^{i1}, m_d^{i2}] - \frac{1}{16} g_2 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} \text{LF}_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^r] + \\ & \frac{1}{16} g_2 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} \text{LF}_{2,2,1,-1} [m_u^r, \tilde{\mu}, m_q^p] + \frac{1}{12} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_d^{i2}] - \\ & \frac{1}{24} g_2 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_q^{i1}] + \frac{1}{8} m_2 \tilde{\mu} g_2^3 y_d^{i1i2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_2, m_q^{i1}] - \\ & \left. \frac{1}{8} g_2 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_q^p, m_u^r] - \frac{1}{8} g_2 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_q^p, m_u^r] \right) \end{aligned}$$